

Learning Path Generator

Problem statement

Learners often know the skill they want to learn but do not know how to break it into manageable stages. Without structure, they jump between resources and topics inefficiently, which slows learning progress. The objective of this project is to build a Generative AI system that takes a target skill or domain as input and generates a structured learning roadmap with stages and topics. The roadmap must remain general, logical, and based only on the provided input.

Expected Output schema

```
{  
  "learning_stages": ["string"],  
  "key_topics": ["string"],  
  "learning_goal_summary": "string"  
}
```

Example input

Data Science

Example output

```
{  
  "learning_stages": [  
    "Beginner",  
    "Intermediate",  
    "Advanced"  
  ],  
  "key_topics": [  
    "Python programming",  
    "Statistics",  
    "Data visualization",  
  ]  
}
```

```
"Machine learning",  
"Model evaluation"  
],  
"learning_goal_summary": "The roadmap introduces the learner to programming and statistics  
first, then progresses toward machine learning concepts and applied model building."  
}
```

Test cases

- Broad skill like Data Science → should create multi-stage roadmap
- Narrow skill like Pandas → should create a focused roadmap
- Very advanced topic → should still produce a logical stage-wise structure rather than random topics